

Advantage

- stepless control of PF
- efficient

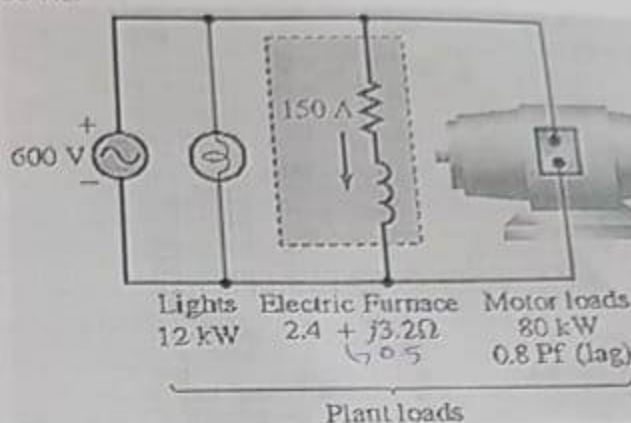
Disadvantage

- maintenance cost
- produce noise

Question 2

(20 Marks)

An industrial client is charged a penalty, if the plant power factor drops below 0.85 lag. The equivalent plant loads are as shown in figure below. The frequency is 60 Hz.



[20]
CLO
02

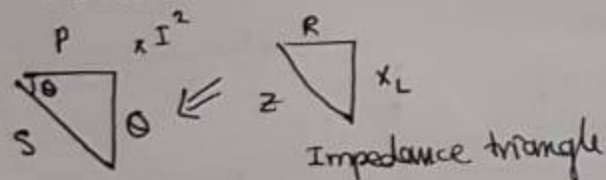
- Determine P_T , Q_T and S [9]
- Determine the value of capacitance (in micro farads) required to bring the power factor up to 0.85 lag. [5]
- Determine generator current before and after correction [4]
- Briefly explain the advantage and a disadvantage of using synchronous condenser over capacitor banks for power factor correction? [2]

Light load 1: $P_1 = 12 \text{ kW}$, $Q_1 = 0 \text{ kVAR}$, $S_1 = 12 \text{ kVA}$

Furnace: $2.4 + j3.2 \Omega$, 150 A

$$P_2 = I^2 R, Q_2 = I^2 X, S = 90 \text{ kVA}$$

$$= 54 \text{ kW}, = 72 \text{ kVAR}$$



Motor: 0.8 PF

$$P_3 = 80 \text{ kW}, Q_3 = S \sin \theta = 60 \text{ kVAR}$$

$$S = \frac{80}{0.8} = 100 \text{ kVA}$$

$$P_T = 12 \text{ k} + 54 \text{ k} + 80 \text{ k} = 146 \text{ kW}$$

$$Q_T = 0 - 72 \text{ k} - 60 \text{ k} = -132 \text{ kVAR}$$

$$S = \sqrt{P_T^2 + Q_T^2} = 196.8 \text{ kVA}$$

② It is operating at

$$\cos \theta_1 = \frac{P_T}{S} = 0.714 \text{ lagging}$$

0.85 lagging (given)

$$Q \text{ of condensers bank} = P(\tan \theta_1 - \tan \theta_2)$$

$$= 146(\tan(\cos^{-1} 0.714) - \tan(\cos^{-1} 0.85))$$

$$= +41.5 \text{ kVAR}$$

$$Q = \frac{V^2}{X_L} \Rightarrow I_C = \frac{V_C}{X_C}, X_C = \frac{1}{2\pi f C}$$

$$I_C = V \times 2\pi f C = 600 \times 2\pi \times 60 \times C$$

$$= 226.19 \times 10^3 C \text{ --- (1)}$$

$$Q = V I_C \sin \theta, \sin \theta = 1,$$

$$I_C = \frac{41.5 \text{ kVAR}}{600} \text{ --- (2)}$$

$$\text{equating (1) \& (2)} \quad C = 305.6 \mu\text{F}$$

③ current before is more & after is less

$\frac{I}{I} \text{ before.}$

$$S = VI$$

$$I = \frac{S}{V}$$

$$= \frac{196.8 \text{ kVA}}{600}$$

$$= 328.63 \text{ A}$$

$$I_{\text{after}}$$

$$S = \frac{P}{\cos \theta} = \frac{146 \text{ k}}{0.85}$$

$$I = \frac{171.7 \text{ k} - 2}{600}$$

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Power Distribution and Utilization	Course Code:	EE 430
Program:	BS (Electrical Engineering)	Semester:	FALL 2017
Duration:	60 Minutes	Total Marks:	40
Paper Date:	18-Sep-17	Weight	15%
Section:	ALL	Pages:	2
Exam:	Midterm-I	Roll No:	14-4519

Instruction/Notes: Answer all questions.

Question 1

(20 Marks)

- How can we improve the diversity factor of a power station? [2]
- If a consumer has connection of five 100-watt lamps and a power point of 500 watts, and a demand factor of 0.8, Find the connected load and maximum load. [2]
- Three turbine-generator units, two rated at 30 MW each and one rated at 10MW are installed to fulfill the following load.

[20]
CLO
01

Time in Hours	6—8	8—12	12—16	16—20	20—24	24—6
Load in MW	20	40	60	20	50	20

Plot the load curve and load duration curve [8].

Calculate/Find:

- Base Load, Peak Load and its duration [2]
- the Energy Generated per day [2]
- Plant Capacity Factor [2]
- Reserve Capacity and Load Factor [2]

Question 4

- a. Draw a single line diagram of a 132 kV power distribution sub-station and network using standard symbols and notations. [8] [5] CLO 05
- b. A substation handles eight 132kV circuits. Four of these circuits are to be implemented in ring bus and other four in double bus single breaker scheme. Draw a typical ring bus scheme and a typical double bus-single breaker scheme for each four circuit system and provide comparison of both schemes in terms of economics and redundancy. [10] [10] CLO 06
- c. A 132kV substation is needed in a metropolitan city. As a power engineer you are responsible for assessing arising issues in its construction and designing. Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice. [5] [5] PLO 06

Question 5

- a. Draw a labeled equivalent circuit of a typical air conditioning compressor and state the power quality problems associated with it. [5] [10] CLO 07
- b. Describe the working/utilization of electrical power by an inverter air conditioner by building electrical circuit model of an inverter AC. Also state advantages (with reasoning) it possess over other conventional air conditioning in terms of design, efficiency, cost, life time, power quality, etc. [5]

Question 6 (Answer briefly)

- a. State reasons why operation of power system at low power factor is undesirable? [2]
- b. Briefly explain why variable nature of load on a power station is a problem for power engineer? [2]
- c. State at least 2 advantages of DC distribution system. [1]
- d. What are the drawbacks of radial distribution scheme? Suggest a better distribution scheme, stating its advantage. [2]
- e. Briefly describe the following terms: [3] [15]
- Load Management
 - Sanctioned load
 - Flicker
- f. Using area-coverage principle, calculate the new load and the area that can be served with the same percent voltage drop, if the voltage from substation is increased 2.5 times. [2]
- g. State first five triplen harmonics. [1]
- h. State five factors which affect the selection of primary-feeder voltage-level. [2]



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Course:	Power Distribution and Utilization
Program:	BS (Electrical Engineering)
Duration:	3 Hours
Paper Date:	27-Dec-17
Section:	ALL
Exam:	FINAL
Answer all questions. Use assumptions where necessary	
Course Code:	EE 430
Semester:	FALL 2017
Total Marks:	85
Weight:	50%
Pages:	2
Roll No:	14-4461

Question 1

- a. It has been desired to install a diesel power station to supply power in a suburban area having the following particulars:

- 1000 houses with average connected load of 1.5 kW in each house. The demand factor and diversity factor being 0.4 and 2.5 respectively.
- 10 factories having overall maximum demand of 90 kW.
- 7 tube-wells of 7 kW each and operating together in the morning.

The diversity factor among the above three types of consumers is 1.5. What should be the minimum capacity of power station? [10]

b. A proposed station has the following daily load cycle :

Time in (Hours)	Load (MW)
6—8	20
8—11	40
11—16	50
16—19	35
19—22	70
22—24	40
24—6	20

Draw the load curve and select suitable generator units from the 10,000, 20,000, 25,000, 30,000 kVA. Assume four sets of each of the above rating. Generators are available. Prepare the operation schedule for the machines selected and determine the load factor from the curve. Assume power factor of 0.9 while scheduling power stations. Also state the criterion used for scheduling. [10]

Question 2

A single phase distributor (2 kilometers long) supplies a load of 120 A at 0.8 p.f. lagging at its far end and a load of 80 A at 0.9 p.f. lagging at its mid-point. Both power factors are referred to the voltage at the far end. The resistance and reactance per km (go and return) are 0.05 Ω and 0.1 Ω respectively. If the voltage at the far end is maintained at 230 V, calculate:

- (i) Voltage at the sending end [8]
- (ii) Phase angle between voltages at the two ends [3]
- (iii) Draw vector diagram of this distributor referred to far end voltage. [4]

Question 3

Draw labeled three phase Open-wye Open-delta connection for transformation from four wire primary to four wire secondary (120/208/240V) service with 210 degree angular displacement. Also state its function/application. [5]

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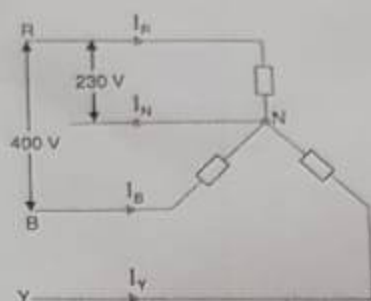


Course:	Power Distribution and Utilization	Course Code:	EE 430
Program:	BS (Electrical Engineering)	Semester:	FALL 2017
Duration:	60 Minutes	Total Marks:	35
Paper Date:	02-Nov-17	Weight:	15%
Section:	ALL	Pages:	1
Exam:	Midterm-II	Roll No:	14-04461

Instruction/Notes: Answer all questions.

Question 1

The three line leads of a 400/230 V, 3-phase, 4-wire supply are designated as R, Y and B respectively. The fourth wire or neutral wire is designated as N. The phase sequence is RYB.



[20]
CLO
3.4/3.5

- Compute the currents in the four wires when the following loads are connected to this supply:
 –From R to N : 30 kW, unity power factor –From Y to N : 43.87 kW, 0.9 lag
 –From B to N : 42.21 kW, 0.866 lead
- Draw phasor diagram to represent voltages and currents in each line.
- If the load from B to N is removed, what will be the value of currents in the four wires?

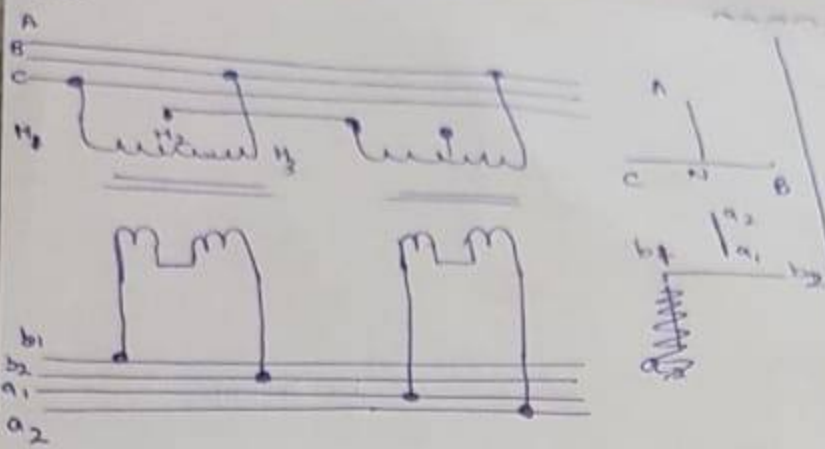
Question 2

Draw labeled T or Scott connection for transformation from three wire primary to four wire secondary service. Also state its function/application.

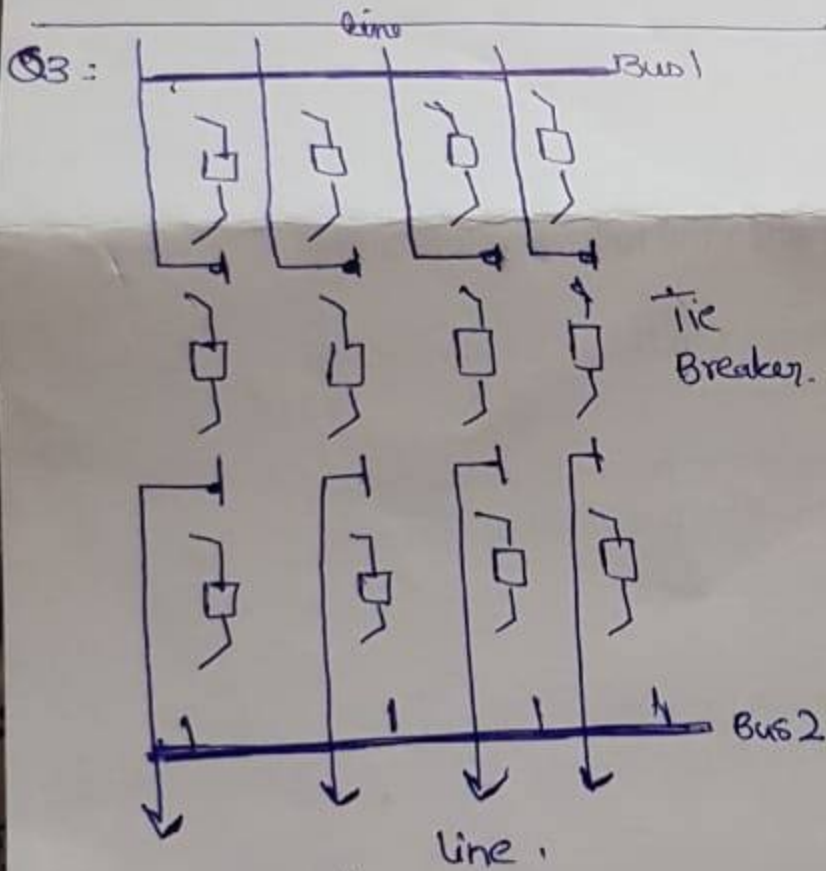
Question 3

Draw breaker and a half scheme and state its advantage over double bus double breaker scheme.

Q2

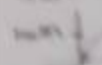


Employs two transformers (main & teaser)
 Use in connection for 3 ϕ to 2 ϕ transformations
 This connection evenly distributes a balanced load b/w the phases of the source.



- High reliable
- Most flexible operation
- Breaker failure of bus side breakers removes only one circuit from service
- Bus failure does not remove any feeder.
- 1 1/2 breakers per circuit.

Q1a. Write formula & by which type of load is the demand factor not exists on the same level.



$$DF = \frac{P_{avg}}{P_{max}} = 0.8$$

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Course: Power Distribution and Utilization
Program: BS (Electrical Engineering)
Duration: 60 Minutes
Paper Date: 18-Sep-17
Section: ALL
Exam: Midterm-I

Course Code: EE 436
Semester: FALL 2017
Total Marks: 45
Weight: 15%
Pages: 2
Roll No: 15-000000

Instruction/Notes: Answer all questions

Question 1

(20 Marks)

- a. How can we improve the diversity factor of a power station? [2]
 b. If a consumer has connection of five 100-watt lamps and a power point of 500 watts, and a demand factor of 0.8, Find the connected load and maximum load. [2] $CL = (5 \times 100) + 500 = 1kW$, $max D = 0.8 \times 1kW = 800W$
 c. Three turbine-generator units, two rated at 30 MW each and one rated at 10MW are installed to fulfill the following load. [20] CLO 01

Time in Hours	6-8	8-12	12-16	16-20	20-24	24-6
Load in MW	20	40	60	20	50	20

Installed capacity = $30 + 30 + 10 = 70 \text{ MW}$

Plot the load curve and load duration curve [8].

Calculate/Find:

Base load

= $20 \text{ MW} - 24 \text{ hours}$

Peak load

= $60 \text{ MW} - 4 \text{ hours}$

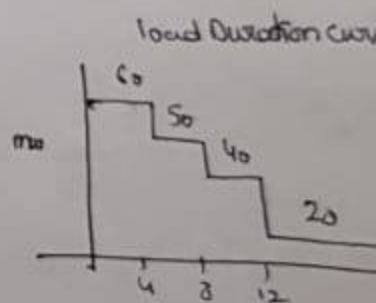
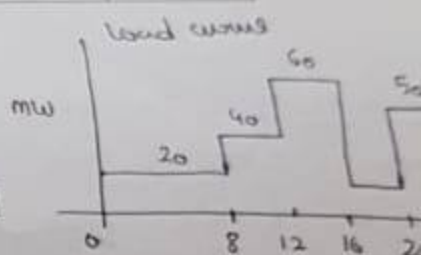
- (i) Base Load, Peak Load and its duration [2]
 (ii) the Energy Generated per day [2]
 (iii) Plant Capacity Factor [2]
 (iv) Reserve Capacity and Load Factor [2]

energy generated per day = $(20 \times 2) + (40 \times 4) + (4 \times 60) + (4 \times 20) + (4 \times 50) + (6 \times 20) = 840 \text{ MWh}$

Plant CF = $\frac{\text{Avg load}}{\text{Plant capacity}} = \frac{\text{unit gen}}{\text{Hme}} = \frac{840 \text{ M}}{24} = 50\%$

RC = $70 \text{ MW} - 60 \text{ MW} = 10 \text{ MW}$

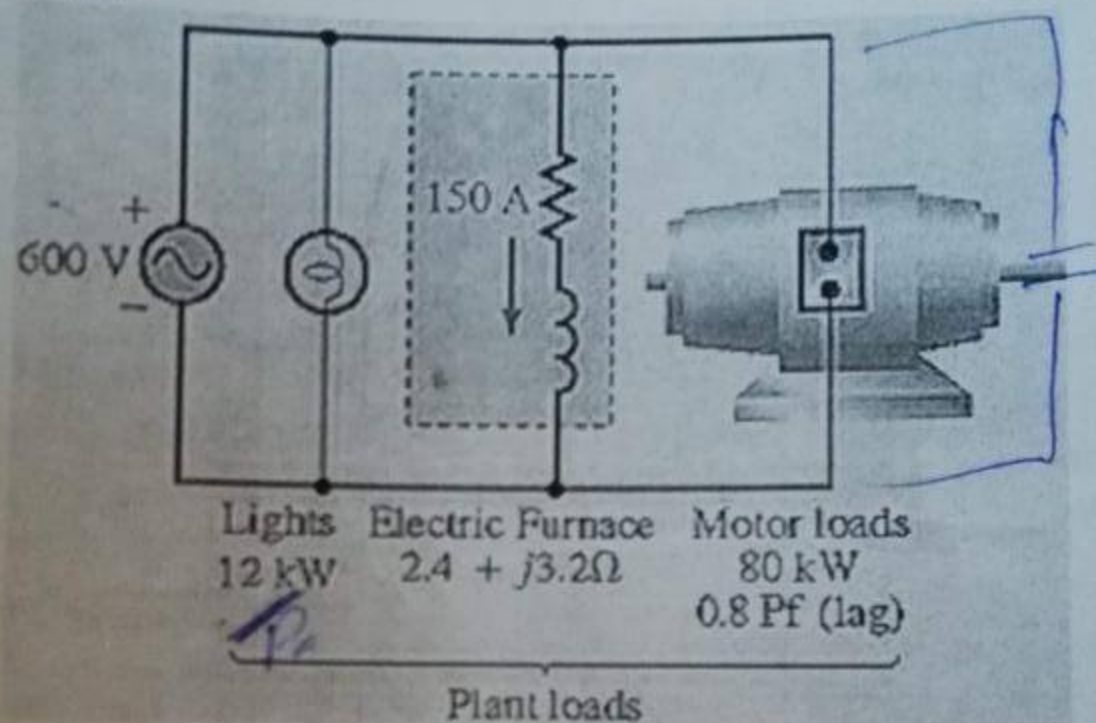
LF = $\frac{\text{Avg load}}{\text{max demand}} = \frac{35}{60} = 0.583$



Question 2

(20 Marks)

An industrial client is charged a penalty, if the plant power factor drops below 0.85 lag. The equivalent plant loads are as shown in figure below. The frequency is 60 Hz.



[20]
CLO
02

- Determine P_T , Q_T and S [9]
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